

Desolvated cations promote CO₂ electroreduction through partial covalent interactions

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Cations play a critical role in electrochemical CO₂ reduction, yet the mechanisms underlying their influence remain incompletely understood owing to limited knowledge of their solvation structures and interfacial interactions under operating conditions. Here we combine surface-sensitive total electron yield X-ray absorption spectroscopy, surface-enhanced infrared absorption spectroscopy and density functional theory calculations to determine the solvation environment of caesium ions near a ligand-modified silver nanocatalyst under CO₂-reducing conditions. We find that desolvated Cs⁺ ions, confined between a detached ligand layer and the silver surface, enable CO₂ activation at potentials as high as 0.4 V versus reversible hydrogen electrode—reducing the overpotential for CO₂-to-CO conversion by 250 mV compared with a bare Ag film. Our results further reveal partial covalent character in Cs–intermediate interactions, which directly accounts for the enhanced catalytic activity. These findings provide multimodal spectroscopic evidence that cations can modulate reaction kinetics beyond purely electrostatic effects.

Renewable energy-powered electrochemical CO₂ reduction (CO₂R) has emerged as a promising approach to mitigate anthropogenic carbon emissions and reduce reliance on fossil fuels by producing fuels and chemical feedstocks^{1,2}. The ultimate goal of CO₂R is to achieve high selectivity towards desired products while minimizing energy input, typically reflected by lower overpotential. In addition to the choice of transition metal electrocatalysts, which are commonly employed for CO₂R, the electrolyte composition—particularly the identity of the

cation—can substantially influence both the energy efficiency and product distribution of the reaction^{3–5}. The effect of cations in CO₂R has been extensively studied, and several key observations have been consistently reported across various catalyst types and cell configurations. First, the presence of alkali-metal cations is essential for the occurrence of CO₂R, particularly in H-cell systems, where the interfacial electric field is relatively weak^{6,7}. Second, CO₂R catalytic activity correlates with the radius of the alkali cation: Cs⁺ > K⁺ > Na⁺ > Li⁺ (refs. 8–11).

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While the critical role of cations in CO₂R is widely recognized, and the trend above has been predominantly observed regardless of catalyst identity, morphology or cell configuration, the underlying mechanisms behind the observed cation-dependent activity trend remain under debate. Broadly, four main hypotheses have been proposed: larger, weakly hydrated cations (for example, Cs⁺) induce a stronger near-surface electric field^{12,13}; decrease the pK_a of cation hydrolysis and thus provide better buffering capacity at the interface^{10,14}; partially shed their solvation shells and stabilize key CO₂R intermediates through electrostatic interactions^{15–18}; and accelerate the hydrogenation of intermediates in acidic conditions⁷. These hypotheses are primarily supported by theoretical simulations, while direct experimental evidence remains limited and often fails to reflect realistic catalytic operating conditions. Furthermore, these explanations are not mutually exclusive—they are essentially different manifestations of the same phenomenon: the solvation properties of cations and their interfacial behaviour under CO₂-reducing conditions. Therefore, to comprehensively explain cation effects in CO₂R, one must combine complementary techniques capable of probing various aspects of cation solvation and track their evolution under operando conditions in real catalytic systems.

Recently, our group developed a catalytic architecture called the nanoparticle-ordered ligand interlayer (NOLI), which exhibits benchmark CO₂ conversion activity^{19,20}. For example, Ag and Au NOLI both achieve 100% Faradaic efficiency (FE) towards CO, at $-0.78 V_{\text{RHE}}$ and $-0.40 V_{\text{RHE}}$, respectively¹⁹; Pd NOLI achieves 100% formate FE at $-0.20 V_{\text{RHE}}$ ¹⁹; and Cu NOLI reaches over 55% FE towards C₂⁺ products at $-0.85 V_{\text{RHE}}$ ²⁰. In the NOLI system, cations are confined in an aqueous layer sandwiched between a detached, ordered ligand layer and the negatively charged metal surface^{19,21}. Notably, once the ligand layer is intentionally removed after NOLI formation, the catalytic performance collapses, despite preservation of nanoparticle size, morphology and crystallinity. This result indicates that the ligand-defined interfacial environment—rather than the metallic scaffold itself—is responsible for the enhanced catalytic performance^{19,22}. Consistent with one of the proposed mechanisms, the ab initio molecular dynamics simulations suggest that cations entering this environment partially shed their solvation shells and stabilize key CO₂R intermediates¹⁹. Importantly, we observe the same cation-dependent trend in NOLI systems across different metallic catalysts, with Cs⁺ consistently enhancing CO₂R the most (Supplementary Fig. 1). Even more intriguingly, for every alkali cation, NOLI shifts the performance profile by approximately 200 mV towards lower overpotentials compared with its metal thin-film counterpart, suggesting that the cation effect is appreciably amplified in NOLI. This makes the NOLI architecture an ideal platform to study cation solvation structures and their relationship with CO₂R activity, as it both preserves and enhances the traditional cation trend—thereby increasing the likelihood of observing solvation structure changes.

In this work, using Ag NOLI as a representative system, we utilized a combination of surface-sensitive vibrational and X-ray spectroscopies to investigate the solvation structure of Cs⁺ and its influence on CO₂R. In situ attenuated total reflection surface-enhanced infrared absorption spectroscopy (ATR-SEIRAS) unveils that at the solid–liquid interface of Ag NOLI, the floating ligand layer maintains a highly confined, ion-rich microenvironment where CO₂ activation is facilitated by

desolvated Cs⁺ at a potential as high as $0.4 V_{\text{RHE}}$, a process that accounts for the superior catalytic activity of NOLI. Concurrently, in situ electron yield X-ray absorption spectroscopy (XAS) at the O K-edge and Cs M-edge demonstrates that the promotional effect of desolvated cations originates from direct interactions with adsorbed *CO₂. Notably, further density functional theory (DFT)-assisted analysis of the Cs M-edge spectra reveals that the Cs–CO₂ interaction exhibits partial covalent character. These findings suggest that Cs⁺ not only stabilizes reaction intermediates electrostatically but also participates in partial orbital hybridization, thereby substantially influencing reaction kinetics. Altogether, this work provides direct spectroscopic evidence for the presence and reactivity of desolvated cations and uncovers a previously underappreciated mechanism of cation–intermediate interaction. By linking distinct mechanisms about cation effects to a unified origin rooted in cation solvation structure, our results establish ligand-enabled interfacial confinement as a generalizable design principle for catalytic systems aiming to leverage cation reactivity.

Results

Ligand-modulated ionic environments facilitate CO₂ adsorption

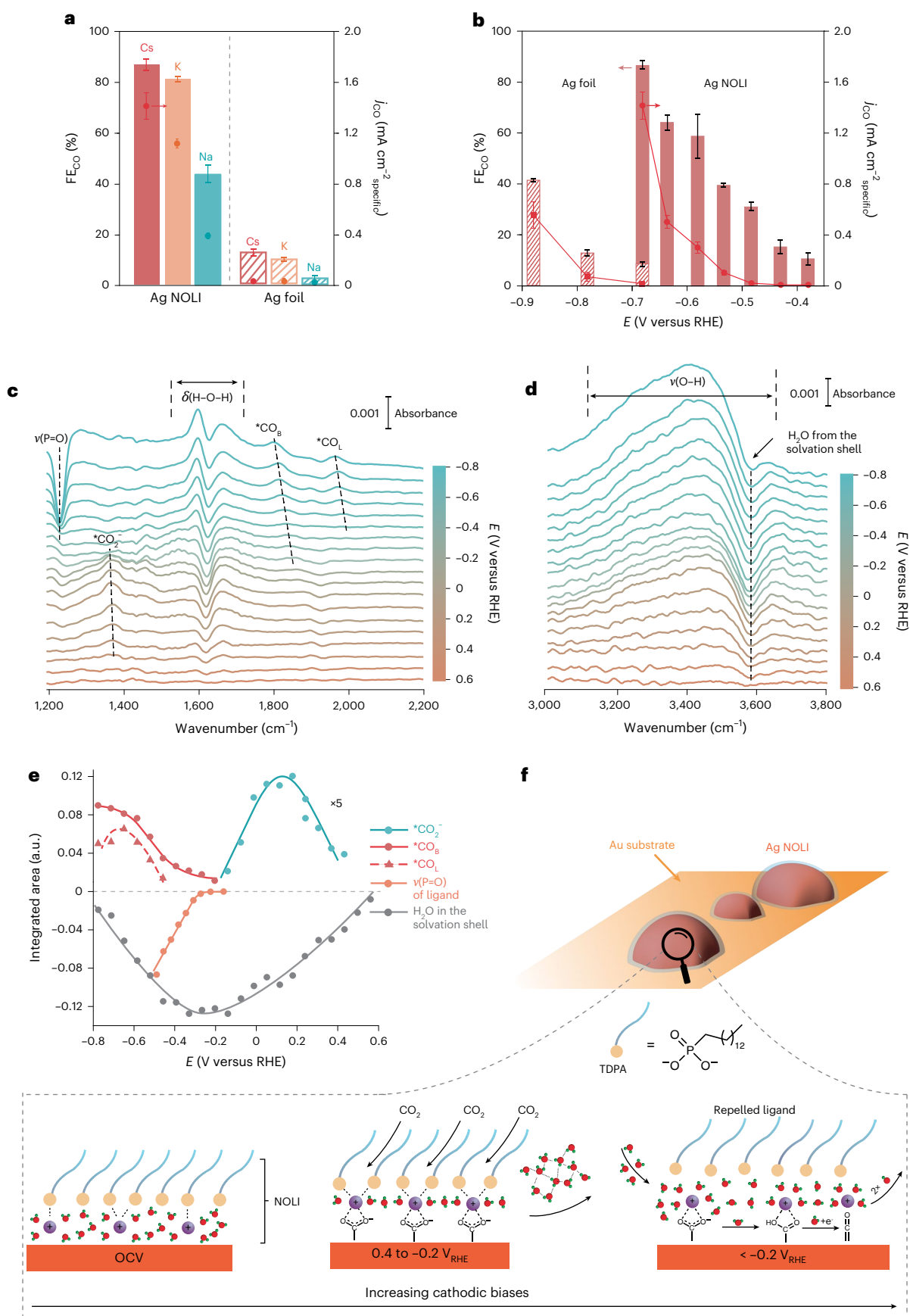
The CO₂R performance of two silver-based systems—Ag foil and Ag NOLI—is shown in Fig. 1a,b. As expected, both systems exhibit a cation-dependent enhancement in activity and selectivity. In the Ag NOLI system, CO FE increases from 42% in 0.1 M NaHCO₃ to 79% in KHCO₃, and further to 85% in CsHCO₃ at $-0.68 V_{\text{RHE}}$. A similar trend is observed for Ag foil, although with much lower absolute values: CO selectivity increases from 2% to 10% and 12% under the same electrolyte conditions. All reported potentials in this work are referenced to the reversible hydrogen electrode (RHE), unless otherwise noted. In both systems, the partial current density for CO production also increases with cation size, from Na⁺ to Cs⁺. Notably, the NOLI system shows a dramatic improvement in both selectivity and activity compared with the Ag foil, regardless of the cation identity. This enhancement highlights the notable promotional effect of the NOLI architecture on CO₂R.

The performance enhancement associated with NOLI is not exclusive to silver. As shown in Supplementary Fig. 1, similar improvements are observed for Cu and Au NOLI systems compared with their corresponding single crystal counterparts, suggesting that the catalytic boost conferred by the NOLI structure is generalizable across different metals. While copper is widely recognized for its unique ability to catalyse the formation of multi-carbon products from CO₂ at appreciable rates, its susceptibility to structural and chemical transformations under electrochemical bias, along with its complex reaction pathways, complicates the study of cation solvation structures^{2,23}. By contrast, silver offers greater stability under applied potentials and features a simpler, better-understood CO₂-to-CO conversion pathway. Therefore, in this work, Ag NOLI was chosen as the model system for studying the cation solvation structure and its influence on CO₂R. We believe that the conclusions drawn from this work may be applied to other heterogeneous electrocatalytic systems.

To elucidate how the NOLI architecture promotes CO₂R at low overpotentials, in situ ATR-SEIRAS measurements were performed using 0.1 M CsHCO₃ as the electrolyte. The schematic of the in situ ATR-SEIRAS cell is shown in Supplementary Fig. 2. Before spectral collection, the

Fig. 1 | CO₂ activation at low overpotential in Ag NOLI. **a**, A comparison of FE_{CO} and partial current density between Ag NOLI and Ag foil, measured at $-0.68 V_{\text{RHE}}$ in CO₂-saturated 0.1 M CsHCO₃, 0.1 M KHCO₃ and 0.1 M NaHCO₃ electrolytes. **b**, FE_{CO} and partial current density as a function of applied potential for Ag NOLI and Ag foil, both tested in CO₂-saturated 0.1 M CsHCO₃. Error bars represent the standard deviation from three independent measurements ($n = 3$), with bars and symbols indicating the mean. **c,d**, In situ ATR-SEIRAS spectra collected from 1,200–2,200 cm^{−1} (**c**) and 3,000–3,800 cm^{−1} (**d**) regions for Ag NOLI. Spectra were acquired under applied potentials ranging from $0.6 V_{\text{RHE}}$ to $-0.8 V_{\text{RHE}}$

in CO₂-saturated 0.1 M CsHCO₃, referenced to a spectrum collected at $0.6 V_{\text{RHE}}$. **e**, The integrated absorbance of interfacial species identified in **c** and **d** as a function of potential (E). Positive areas are magnified fivefold for clarity. **f**, A schematic of interfacial processes occurring at different biases for Ag NOLI. The TDPA ligand is depicted as a light-yellow ball (phosphonic acid headgroup) linked to a cyan chain (14-carbon alkyl chain). Cs⁺ ions are shown as purple spheres with a central plus sign. Water molecules are represented by a red sphere (oxygen) and two green spheres (hydrogen). For clarity, species outside the ligand layer and minor components (for example, anions) are omitted.



electrode was preconditioned by a linear sweep voltammetry from open-circuit potential (OCV; around $0.6\text{ V}_{\text{RHE}}$) to $-0.8\text{ V}_{\text{RHE}}$, followed by stabilization at $-0.8\text{ V}_{\text{RHE}}$ for at least 10 min. This preconditioning step induces ligand detachment and the aggregation of small Ag nanoparticles into larger polycrystalline domains, establishing the NOLI structure (Supplementary Figs. 3 and 4). A detailed discussion of changes of the interfacial microenvironment during this preconditioning process is provided in Supplementary Note 1. The successful formation of NOLI was confirmed by observing the dynamic attraction/repulsion behaviour of the ligands in response to applied bias (Supplementary Fig. 6).

After NOLI formation, the electrode was rested at OCV for 30 min under continuous CO_2 bubbling to remove surface-adsorbed species formed during preconditioning. The infrared (IR) spectrum at this state was used as a reference for all subsequent spectra. Figure 1c,d presents SEIRAS spectra of the fingerprint region ($1,200\text{--}2,200\text{ cm}^{-1}$) and water stretch region ($3,000\text{--}3,800\text{ cm}^{-1}$) under increasingly negative biases. A feature at $1,372\text{ cm}^{-1}$ emerges around $0.4\text{ V}_{\text{RHE}}$, reaches maximum intensity near $0.1\text{ V}_{\text{RHE}}$ and then gradually decreases with further negative bias, disappearing entirely around $-0.2\text{ V}_{\text{RHE}}$, while shifting slightly to $1,362\text{ cm}^{-1}$. Following the disappearance of this feature, a new peak around $1,846\text{ cm}^{-1}$ appears, accompanied by an increasingly negative band at $1,232\text{ cm}^{-1}$. On the basis of our previous work²¹, the $1,232\text{ cm}^{-1}$ band is assigned to the $\nu(\text{P}=\text{O})$ vibration of tetradecylphosphonic acid (TDPA) ligands (Supplementary Note 1), where the negative signal indicates ligands moving further away from the Ag surface.

In addition, a peak around $1,990\text{ cm}^{-1}$ is observed from $-0.4\text{ V}_{\text{RHE}}$ to $-0.8\text{ V}_{\text{RHE}}$, shifting to lower wave numbers as the bias becomes more negative. By -0.8 V , the $1,846\text{ cm}^{-1}$ and $1,990\text{ cm}^{-1}$ features shift to $1,804\text{ cm}^{-1}$ and $1,966\text{ cm}^{-1}$, respectively. The observed Stark shifting behaviour strongly suggests that these modes correspond to surface-adsorbed species. We assign these two modes to bridge-adsorbed ($^*\text{CO}_\text{b}$) and linearly adsorbed ($^*\text{CO}_\text{l}$) CO species, respectively, which aligns well with literature reported IR frequencies of CO adsorbed on Ag surfaces^{24–26}. This assignment is further supported by the observation that these bands increase in intensity over time when CO_2 is reintroduced into a N_2 -saturated 0.1 M CsHCO_3 electrolyte at $-0.8\text{ V}_{\text{RHE}}$ (Supplementary Fig. 7), indicating that they are intermediates of CO_2 R rather than surface-bound hydrogen, which has also been reported in this spectral window²⁷. Notably, bridge-bonded $^*\text{CO}$ is typically favoured at low CO coverage on Ag, while $^*\text{CO}_\text{l}$ becomes dominant at higher coverage²⁸, consistent with our observation that the $1,846\text{ cm}^{-1}$ feature appears first at lower overpotentials. This behaviour rules out the possibility of $^*\text{CO}$ adsorbed on Pt—a common source of contamination in SEIRAS—since Pt favours atop-bound CO at low coverages^{29–31}. Although both bridge- and linear-bound CO species are observed, CO-induced site blocking is unlikely under the present conditions. CO is generated only in situ at very low partial pressures ($\sim 6 \times 10^{-4}\text{ atm}$), far below the regime where competitive CO/ CO_2 adsorption has been reported³².

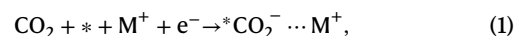
The nature of the $1,372\text{ cm}^{-1}$ feature was further investigated. Given the potential-dependent shift of this vibrational mode, it is most probably associated with a surface-bound species. While vibrational modes associated with (bi)carbonate species are commonly observed in this spectral region³³, such an assignment appears unlikely in our system. Adsorbed $^*\text{HCO}_3^- / ^*\text{CO}_3^{2-}$ would be expected to detach under increasingly negative potentials, leading to a progressively negative peak rather than the observed buildup and decay. To probe whether this feature involves hydrogen, ATR-SEIRAS measurements were repeated in 0.1 M CsDCO_3 (in D_2O), as vibrational modes involving hydrogen typically experience red-shift when hydrogen is replaced by deuterium⁷. No noticeable shift was observed in the D_2O spectra (Supplementary Fig. 8), indicating that the $1,362\text{ cm}^{-1}$ mode does not involve hydrogen. Given that the appearance and disappearance of this feature coincide with the emergence of $^*\text{CO}$ adsorption, we propose that the $1,362\text{ cm}^{-1}$ band corresponds to the symmetric O–C–O stretch ($\nu_s(\text{O}=\text{C}=\text{O})$) of an adsorbed CO_2 intermediate. The asymmetric stretch,

typically expected around $1,500\text{ cm}^{-1}$, is not observed owing to the surface selection rules of SEIRAS^{7,34,35}. Importantly, adsorption of CO_2 on metal surfaces requires activation of the linear molecule through bending, which lifts the degeneracy of the $2\pi_u$ lowest unoccupied molecular orbital and enables hybridization with metal states. This interaction necessarily involves partial electron transfer from the metal into the CO_2 antibonding orbital, yielding a bent adsorbate with partial negative charge^{6,17}. Therefore, the observed $1,372\text{ cm}^{-1}$ feature is less likely to correspond to a charge-neutral adsorbed CO_2 species. Instead, we assign it to (partially) reduced CO_2 , denoted by $^*\text{CO}_2^-$ or activated CO_2 in the following discussion.

In addition to carbon-related intermediates, changes in water-related modes were also observed. In the fingerprint region (Fig. 1c), the H–O–H bending mode shows a negative feature around $1,620\text{ cm}^{-1}$ that becomes more prominent under negative biases. New features at $1,600\text{ cm}^{-1}$ and $1,660\text{ cm}^{-1}$ gradually build up as the potential becomes more negative. In the OH stretching region (Fig. 1d), the negative band at $3,580\text{ cm}^{-1}$ becomes more intense at moderate biases and then partially recovers at higher negative potentials, accompanied by intensity increases at $3,200\text{--}3,400\text{ cm}^{-1}$. Similar effects are reproduced in the D_2O stretching mode in 0.1 M CsDCO_3 (Supplementary Fig. 9). The synchronous evolution of the $1,620$ and $3,580\text{ cm}^{-1}$ features suggests they belong to the same water species. Notably, the $1,620$ and $3,580\text{ cm}^{-1}$ features have been observed in ATR-IR spectra of highly concentrated electrolytes ($\sim 10\text{ M}$), where their intensity scales with ion concentration (Supplementary Fig. 10). Therefore, these bands are assigned to water molecules tightly associated with the solvation shell of cations, which is consistent with multiple spectroscopic studies on interfacial water structure^{36–38}.

The integrated band intensities of the vibrational modes detected in Fig. 1c,d at different applied potentials are summarized in Fig. 1e. In the potential window from OCV to $-0.2\text{ V}_{\text{RHE}}$, the increasing negative area associated with H_2O in the cation solvation shell with increasingly negative potentials indicates progressive cation desolvation, that is, the reduction of the number of water molecules coordinated to the cation. Interestingly, activated CO_2 is also observed within this same potential window, notably more positive than the potentials typically reported for $^*\text{CO}_2$ activation on Cu (refs. 7,39). The simultaneous occurrence of these two phenomena suggests that CO_2 activation at such low overpotentials is facilitated by desolvated cations. As illustrated schematically in Fig. 1f, within this potential range, $^*\text{CO}_2$ molecules replace the H_2O molecules originally present in the cation solvation shell. The displaced water molecules reorganize into a bulk-like phase with a strong hydrogen-bond network, which explains the intensity increase observed between $3,200$ and $3,400\text{ cm}^{-1}$ in Fig. 1d—a spectral signature characteristic of strongly hydrogen-bonded, bulk-like water.

Notably, $^*\text{CO}$ species only appear when the applied potential becomes more negative than $-0.2\text{ V}_{\text{RHE}}$, coinciding with the disappearance of the $^*\text{CO}_2^-$ signal. This observation suggests that the rate-determining step (RDS) for CO formation is the generation of $^*\text{CO}_2^-$ via the first electron-transfer activation step, rather than the subsequent conversion of $^*\text{CO}_2^-$ to $^*\text{CO}$. Interestingly, the emergence of $^*\text{CO}$ also coincides with the recovery of water-related vibrational bands (Fig. 1e), which might initially imply that protons or water molecules participate in the RDS. However, as shown in Supplementary Fig. 11, the CO partial current density (i_{CO}) is nearly identical in both deuterated and protic electrolytes, indicating that proton transfer does not contribute to the RDS. Together, these findings support a mechanism in which CO_2 activation proceeds via an electron-transfer-coupled adsorption step facilitated by desolvated cations



which constitutes the RDS for CO formation. The observed recovery of water-related signals is therefore attributed to further ligand

detachment at $-0.2 V_{\text{RHE}}$ or more negative potentials, which enables increased water access to the interfacial region, rather than indicating a direct role of water in the RDS of *CO formation.

The production of gaseous CO on Ag NOLI is first detected around $-0.4 V_{\text{RHE}}$ (Fig. 1b), a potential where *CO_L starts to be observed, suggesting that a critical surface coverage of CO must be reached before desorption occurs. To elucidate the origin of the promotion effect of CO_2 activation in the NOLI architecture, SEIRAS spectra were also collected for a ligand-free Ag film. As shown in Supplementary Fig. 12, under negative bias, the negative band associated with H_2O stretching is nearly absent, indicating that the interfacial cations remain more solvated than in the NOLI system under equivalent potentials. Moreover, no discernible *CO_2 -related features appear across the scanned potential range, probably due to lower $^*CO_2^-$ surface coverage compared with Ag NOLI. Consequently, the onset potentials for both *CO_B and *CO_L species on the Ag film are at least 250 mV more negative than those observed in Ag NOLI. This comparison highlights that CO_2 activation is substantially facilitated by desolvated cations within the NOLI architecture.

Results from in situ SEIRAS clearly reveal the promotional effect of desolvated cations on *CO_2 activation in Ag NOLI. To determine whether this enhancement arises primarily from an enhanced interfacial electric field generated by desolvated cations or from direct stabilization of reaction intermediates, we probed the Stark tuning behaviour of adsorbed *CO_L to assess the interfacial electric field of Ag NOLI (see Supplementary Note 2 for details). As shown in Supplementary Fig. 13, within the potential range where $^*CO_2^-$ is observed ($0.4 V_{\text{RHE}}$ to $-0.2 V_{\text{RHE}}$), the vibrational Stark tuning rate of *CO_L is approximately $8 \text{ cm}^{-1} \text{ V}^{-1}$, which is substantially smaller than the $-30 \text{ cm}^{-1} \text{ V}^{-1}$ typically reported for *CO on metal films^{8,39–41}. This comparison indicates that the local electric field in this potential window is relatively modest and enables us to exclude mechanisms solely on the basis of amplified interfacial electric fields. Instead, these results point towards a mechanism in which direct interactions between desolvated Cs^+ ions and adsorbed $^*CO_2^-$ intermediates play a decisive role in promoting this activation—a hypothesis that will be further examined in the following sections.

Interactions within the water-deficient NOLI pocket

To elucidate the interactions between desolvated Cs^+ and interfacial species in Ag NOLI, in situ XAS was employed. Interface-sensitive information was obtained using total electron yield (TEY) detection. Figure 2a schematically illustrates the liquid flow cell configuration used for the XAS measurements in this study. In this liquid-cell geometry, X-ray absorption at the solid–liquid interface creates a core hole. Core-hole relaxation generates Auger electrons, whose kinetic energies are determined by atomic energy levels. These Auger electrons undergo inelastic scattering in the surrounding aqueous medium and generate cascades of low-energy secondary electrons⁴². Owing to the short mean free path of these electrons in aqueous media, only those generated within a few nanometres of the electrode can reach the conductive surface and be collected as a photoinduced current before recombining with ionized species in the electrolyte^{43,44}. This photoinduced current (the TEY signal) follows the X-ray absorption cross-section and is used to reconstruct the XAS spectra. Further experimental details are provided in the Methods.

We first focus on the oxygen K-edge XAS spectra, as it is well-established for characterizing aqueous systems^{45,46}. In condensed phases, water typically exhibits three distinct spectral features: a pre-edge near 535.0 eV, a main edge near 537.5 eV and a post-edge near 541.0 eV. The pre-edge intensity scales with the asymmetry in the local molecular environment, while the post-edge scales with the number of H bonds^{47–50}. For example, liquid water displays a strong pre-edge and subtle post-edge, while ice shows a more subtle pre-edge and more pronounced post-edge. As shown in Fig. 2b, the interfacial

water structure near a thin silver film exhibits minimal changes from pure water to 0.5 M $CsHCO_3$; only a slight increase in the pre-edge intensity is observed, probably due to the structure-breaking nature of Cs^+ cations¹⁰. By contrast, Ag NOLI displays markedly different behaviour (Fig. 2c). Before preconditioning, when ligands are covalently attached to the Ag surface, water is excluded from the surface by the hydrophobic alkyl chains of the ligands. In this initial state, the phosphonic acid headgroup is expected to be the only oxygen-containing species detectable near the surface (Supplementary Fig. 3). Indeed, the absence of interfacial water at OCV is evident from the spectrum, which shows a strong peak at 535.2 eV and a weaker second peak at 536.9 eV, an intensity trend that is opposite to that of water. We assign this spectrum to TDPA molecules chemisorbed onto the Ag nanoparticle surface via a bidentate configuration²¹. Dry TDPA powder exhibits a similar spectrum; minor differences can be ascribed to the different protonation state of the headgroup and solvation effects⁵¹.

After applying a bias of $-0.8 V_{\text{RHE}}$ for 10 min to induce NOLI formation and then returning to OCV (around $0.6 V_{\text{RHE}}$), the oxygen K-edge XAS spectrum changes drastically to resemble that of water, indicating the formation of an aqueous pocket at the interface (Fig. 2c) caused by preconditioning. Upon applying a potential of $0.4 V_{\text{RHE}}$ —close to the onset of $^*CO_2^-$ adsorption as shown in Fig. 1e—the O K-edge spectrum undergoes another dramatic shift, reverting to a shape similar to that of pure TDPA. Almost no intensity remains near water's main and post-edge features, indicating that a considerable amount of the water previously confined within the NOLI pockets at OCV has been expelled further from the surface at $0.4 V_{\text{RHE}}$, leading to the formation of a highly water-deficient environment populated by desolvated Cs^+ cations. This observation is consistent with SEIRAS measurements, where the OH stretching bands show negative features under $0.4 V_{\text{RHE}}$. The highly ionic environment decreases the Debye length and rapidly screens the negative charge at the interface, enabling the ligands to settle closer to the surface. Compared with chemisorbed TDPA (the OCV spectrum before preconditioning), the peak maximum at $0.4 V_{\text{RHE}}$ appears slightly red-shifted. A similar red-shift has been observed in first-principles XAS simulations for the protonation of sulfate species⁵², leading us to propose that protonation of the phosphonic acid headgroup following ligand detachment is responsible for the red-shift observed here²¹, although contributions from interactions with other interfacial species cannot be ruled out.

When applying an even more negative bias of $0.2 V_{\text{RHE}}$, a more water-deficient spectrum is obtained. Interestingly, the peak maximum continues to red-shift, and a new broad pre-peak emerges at $\sim 528.8 \text{ eV}$ —a feature that is absent in both pure TDPA and pure water spectra. To further investigate this signature, we prepared a Cs –TDPA compound where oxygen atoms of the phosphonic acid headgroup are interacting with a Cs cation (Fig. 2c). The spectrum of this compound matches well with that of the $0.2 V_{\text{RHE}}$ spectrum, with a pronounced shoulder around 528.8 eV (Fig. 2c). This finding suggests a direct interaction between the TDPA headgroup and Cs^+ at $0.2 V_{\text{RHE}}$.

Together, the in situ XAS and ATR-SEIRAS results demonstrate that under moderate bias, the NOLI pocket becomes a highly confined, water-deficient interfacial environment dominated by desolvated cations, detached ligands and surface intermediates. In this micro-environment, Cs^+ not only sheds its hydration shell but also directly interacts with detached ligands. However, O K-edge spectra alone do not provide direct information on Cs –intermediate interactions. To probe these interactions, we next turned to in situ XAS at the Cs M-edge.

Interpretation of Cs M-edge spectra

Recent work has demonstrated that in situ XAS at alkali-metal edges can directly capture electronic structure changes of cations induced by interactions with adsorbed reaction intermediates⁵³, motivating the use of Cs M-edge XAS to directly interrogate Cs –intermediate coupling under operando CO_2R conditions. To develop a fundamental

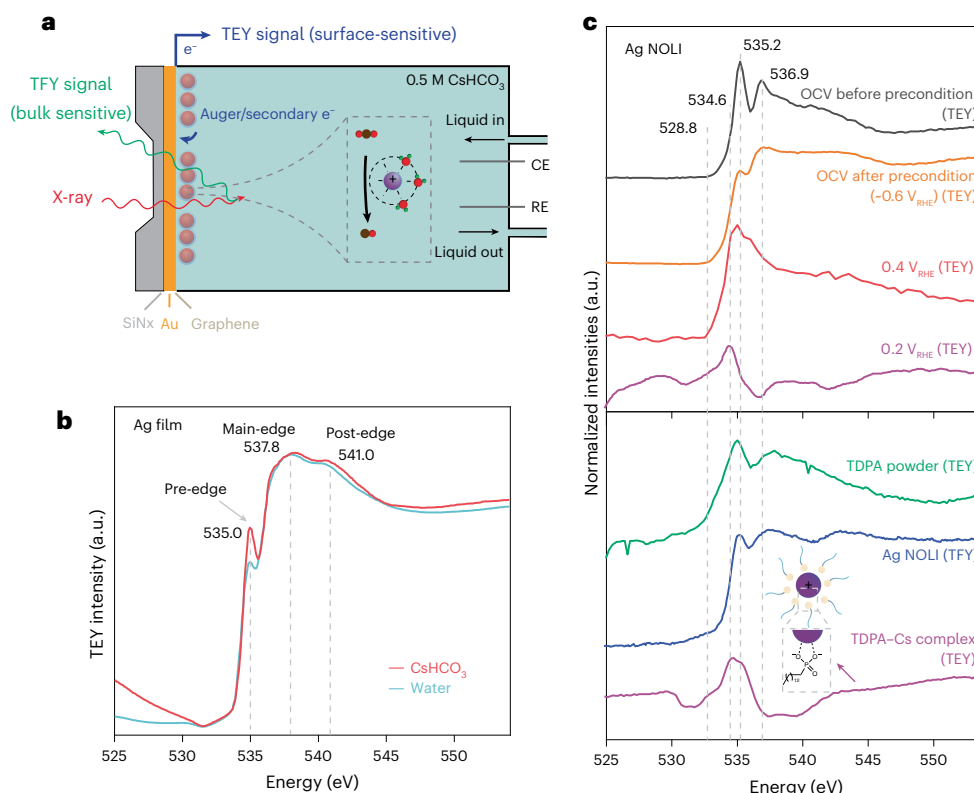


Fig. 2 | O K-edge spectra measured by TEY-XAS. a, A schematic of the custom-built liquid flow X-ray cell. A silicon nitride window is coated with thin layers of chromium and gold to ensure good electrical conductivity. Three layers of graphene are subsequently deposited onto the gold to prevent electronic interference from the metal substrate. The catalyst is then dropcast or coated onto the graphene surface. Measurements are performed under continuous liquid flow, with a 3 M Ag/AgCl electrode and a platinum wire serving as the

reference electrode (RE) and counter electrode (CE), respectively. **b, c**, The electron yield O K-edge spectra of Ag film in H₂O and CO₂-saturated 0.5 M CsHCO₃ (**b**), Ag NOLI at different applied potentials in CO₂-saturated 0.5 M CsHCO₃ (**c**) and reference samples. The reference spectra include TDPA powder (green trace) and Cs-TDPA complexes dissolved in ethanol (red trace, with the solvent background subtracted). Inset: schematic of the Cs-TDPA complex, showing Cs coordinated by the O atoms of TDPA.

understanding of the behaviour of the largely unexplored Cs M-edges, we first measured several relevant reference samples. Figure 3a shows the Cs M₄ and M₅ edges of solid-state Cs₂CO₃, CsHCO₃ and CsI. These absorption features arise from electronic transitions from the 3d core level to the unoccupied 4f orbitals, with the two main absorption edges resulting from spin-orbit coupling of the 3d core hole. Interestingly, additional satellite features are observed at relatively higher energies—specifically at -742.5 eV and -757.0 eV. The intensity of these satellite features is highest in Cs₂CO₃, weaker in CsHCO₃ and barely visible in CsI.

A similar trend has been reported by Suzuki et al., who observed prominent satellite structures in vermiculite (a mineral containing Cs and O), but much weaker features in caesium halides⁵⁴. They attributed the appearance of these satellites to the formation of delocalized 4f orbitals via hybridization between Cs 4f and anion orbitals. While Cs 4f orbitals exhibit minimal overlap with halide orbitals, they strongly interact with O 2p orbitals, explaining the more pronounced tail structures in oxygen-containing compounds. Notably, comparable behaviour has also been observed in M-edge spectra of lanthanide oxides, where similar satellite features arise from hybridization between lanthanide 4f orbitals and O 2p orbitals⁵⁵. Together, these findings suggest that the satellite intensities in our Cs M-edge spectra reflect the degree of orbital mixing between Cs 4f and neighbouring anion orbitals. In addition, they also indicate that beyond the pure electrostatic effect, Cs and O form a partial covalent bond in CsHCO₃ and Cs₂CO₃.

Although the M-edge tail structure provides insight into the local coordination environment of the unoccupied 4f orbitals, the 4f orbitals in Cs⁺ are largely empty in the ground state. To assess whether Cs M-edge features reflect broader ground-state interactions, we

therefore combined density of states (DOS), Bader charge analysis, differential charge density ($\Delta\rho$) and electron localization function (ELF) calculations for Cs₂CO₃, CsHCO₃ and CsI. As shown in Fig. 3b, the DOS reveals that Cs 5s and 5p orbitals lie within the energy ranges of -20 to -25 eV and -5 to -10 eV, respectively. In CsI, no orbital overlap is observed, and orbitals show no dispersion, indicating purely ionic interactions between Cs and I. By contrast, both Cs₂CO₃ and CsHCO₃ show clear overlap between Cs 5p and O 2p orbitals, with a noticeable dispersion of the Cs 5p DOS with a width of more than 2.5 eV, indicating partial covalent bonding between Cs and O. This is further supported by Bader charge analysis, differential charge density and ELF in Fig. 3c: in CsI, the Cs Bader charge approaches its formal ionic value, indicating nearly complete charge separation. In CsHCO₃ the Cs positive charge is slightly reduced and even more so in Cs₂CO₃, indicating partial electron density redistribution towards Cs. Carbonate and bicarbonate phases display directional electron depletion near O and accumulation along the Cs-O axis and partially towards Cs, forming a polarized bridge region; CsI lacks any continuous accumulation between Cs and I, retaining spherical, isolated ionic basins. For ELF slices, along Cs-I, the internuclear gap maintains a uniformly low ELF baseline (<0.30) with no mid-segment basin. In Cs₂CO₃ and CsHCO₃, after excluding near-nuclear regions, an extended intermediate ELF plateau (-0.30–0.35) develops along the Cs-O corridor; the corridor area fraction with ELF >0.30 is markedly larger (Cs₂CO₃ approximately 0.460, CsHCO₃ approximately 0.419 and CsI approximately 0.000) (Fig. 3d, Supplementary Note 3 and Supplementary Table 1). This plateau is well below the high ELF values (-0.7–0.9) characteristic of strong covalent bonds, but it reveals directional polarization and weak

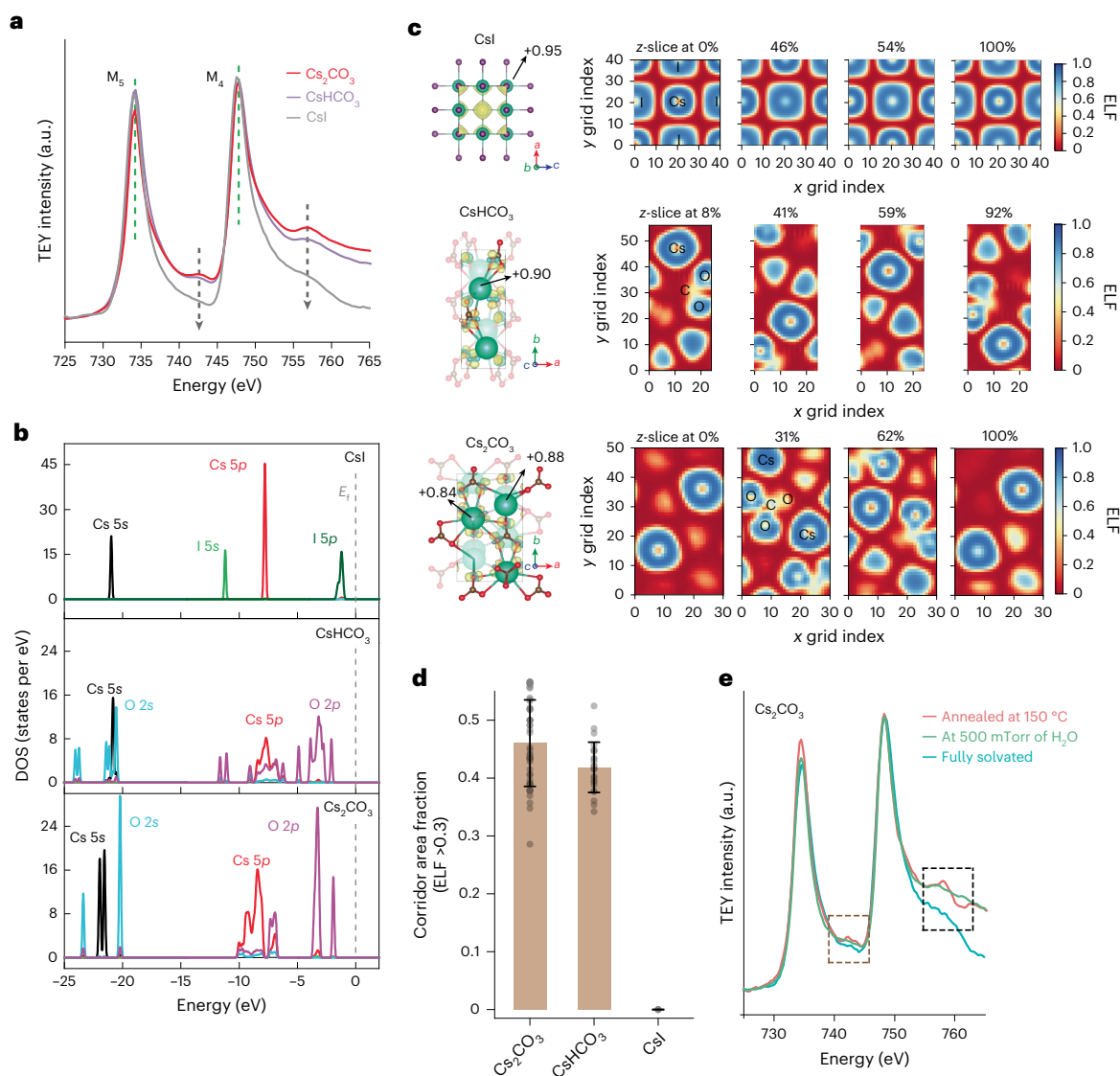


Fig. 3 | Cs M-edge reference spectra. **a**, Cs M-edge spectra of solid-state Cs_2CO_3 , CsHCO_3 and CsI . **b**, The PDOS highlighting Cs 5s and 5p states and anion valence states: I 5s/5p in CsI (top), and O 2s/2p in CsHCO_3 (middle) and Cs_2CO_3 (bottom). The Fermi level is set to 0 eV. Carbonate phases show a substantial orbital mixing between Cs 5p and O 2p states, whereas CsI exhibits well separated ionic manifolds. **c**, The differential charge density ($\Delta\rho = \rho_{\text{total}} - \sum \rho_{\text{frag}}$ (identical supercells, fragment geometries frozen) isosurfaces (positive accumulation/negative depletion) superimposed on the crystal structures (view along the x–y plane) for CsI , CsHCO_3 and Cs_2CO_3 , including Bader charge values on Cs atoms, revealing directional Cs–O polarization absent in Cs–I. Corresponding ELF slices

at four representative z positions, illustrating the emergence of intermediate internuclear charge density corridors for Cs–O but not Cs–I. Atom colours: Cs (green), I (purple), O (red), C (brown) and H (white). **d**, The corridor area fraction with $\text{ELF} > 0.3$ for Cs_2CO_3 , CsHCO_3 and CsI . Each point represents one Cs–anion pair environment in the calculated unit cell, with $n = 48$ Cs–O pairs for Cs_2CO_3 , $n = 24$ Cs–O pairs for CsHCO_3 and $n = 4$ Cs–I pairs for CsI . Bars and error bars show mean values $\pm$ s.d. across Cs–anion pair environments. **e**, Cs M-edge spectra of Cs_2CO_3 measured under different water concentrations. Regions of 738–746 eV and 754–762 eV are highlighted by dashed rectangles, reflecting the evolution of satellite features with increasing water content.

p–*p* co-localization—a polarized partial-covalent character unique to Cs–O in these lattices. Taken together, the substantial Cs 5p–O 2p DOS overlap and dispersion, the dipolar $\Delta\rho$ redistribution, the reduced (less ionic) Bader charge on Cs and the intermediate ELF plateau collectively rationalize the presence of the M-edge satellite tail in Cs_2CO_3 and CsHCO_3 , while the absence of all four signatures in CsI explains the lack of a comparable tail.

The difference in tail intensity between CsHCO_3 and Cs_2CO_3 in the M-edge spectra correlates with the aggregate degree of polarized Cs–O bonding rather than a large increase in per-bond covalency. Bader charge analysis reveals that Cs ions in Cs_2CO_3 exhibit a slightly lower charge (+0.84 and +0.88) than the Cs ion in CsHCO_3 (+0.90), indicating a more pronounced charge redistribution (partial electron

back-polarization) for Cs_2CO_3 . Atomic bond population analysis with the integrated Crystal Orbital Hamiltonian Population (ICOHP) metrics further support this view: although the average per-bond ICOHP for strong Cs–O contacts (2.8–3.4 Å subset) is similar in the two phases (–0.170 versus –0.177 eV per bond; a negative ICOHP indicating a bonding interaction), Cs_2CO_3 hosts a substantially larger number of such interactions (26 versus 16) in the unit cell, yielding a more negative cumulative ICOHP (–4.41 versus –2.83 eV). When all identified Cs–O contacts are included, the total ICOHP in Cs_2CO_3 (–5.54 eV over 60 bonds) also exceeds that in CsHCO_3 (–3.28 eV over 28 bonds), reflecting a denser network of weak-to-moderate polarized bonds. Consistent with this, the Cs–O distance distribution for Cs_2CO_3 is shifted towards shorter separations (Supplementary Fig. 14b,d), increasing

orbital overlap pathways for final-state screening. Taken together, these results indicate that the more extensive network of polarized partial-covalent Cs–O contacts in Cs_2CO_3 —rather than a dramatic strengthening of any single bond—underpins its stronger M-edge satellite tail. Collectively, our experimental and computational results demonstrate that the intensity of satellite features in Cs M-edge spectra can serve as a sensitive probe of the strength and nature of interactions between Cs and its surrounding atoms.

To establish a baseline for interpreting in situ Cs M-edge spectra under electrochemical conditions—where cations are present in a solvated state—we further investigated the spectral evolution of Cs M-edges during the dissolution process. As shown in Fig. 3e, Cs_2CO_3 initially dried under vacuum calcination exhibits sharp satellite features at the previously identified energies, reflecting strong and well-defined Cs–O interactions in the crystalline lattice. However, upon gradual introduction of water vapour, these satellite features progressively broaden and eventually disappear at a water pressure of approximately 500 mTorr. The presence of water is expected to expand and/or distort the crystal lattice, leading to partial solvation. Once Cs^+ ions become fully solvated, a marked decrease in the tail intensity of the M-edge spectra is observed, indicating weakened interactions between Cs^+ and its surrounding oxygen atoms. This weakening is consistent with the longer Cs–O distances in solvation shells compared with the short, directional Cs–O bonds in solid-state salts (Supplementary Fig. 14). These results confirm that M-edge satellite intensity is sensitive not only to the chemical identity of neighbouring atoms but also to the local solvation structure, providing a valuable reference point for interpreting in situ measurements under electrochemical conditions.

The Cs–O interaction in the NOLI pocket

Building on our understanding of Cs M-edge spectral features, we next investigated the interaction between Cs^+ and various interfacial species within the Ag NOLI pocket using in situ Cs M-edge XAS. As outlined in Fig. 4a, Cs M-edge spectra were first collected at OCV, followed by catalyst preconditioning at $-0.2 V_{\text{RHE}}$. After preconditioning, the potential was returned to OCV, and Cs M-edge spectra were recorded every 20 min to monitor temporal changes. It is important to note that directly measuring Cs M-edge TEY signals under $-0.2 V_{\text{RHE}}$ is challenging. The large Faradaic current—typically on the order of microamperes in our setup—overwhelms the TEY signal, which is typically in the nanoampere range. Therefore, while the post-bias spectra may not perfectly capture the $-0.2 V_{\text{RHE}}$ condition, their time-resolved evolution still offers valuable insights into interfacial Cs^+ behaviour during catalysis. This is especially informative in the NOLI system, where the floating ligand layer restricts the rapid relaxation of the electric double layer once the bias is removed.

Before preconditioning, when the surfaces of Ag nanoparticles are chemisorbed with TDPA ligands, the Cs M-edge signal is weak in TEY mode, as shown in Fig. 4b. This is because Cs^+ ions reside outside the ligand layer at this stage, and the electrons emitted from these ions cannot reach the electrode owing to the short mean free path. After applying $-0.2 V_{\text{RHE}}$, an overall enhancement in the TEY signal is observed, indicating an increased local Cs^+ concentration at the electrode interface (Supplementary Fig. 3). Concurrent O K-edge and SEIRAS measurements confirm reduced interfacial water content under bias, implying that Cs^+ ions at the interface are partially desolvated. Interestingly, at 0 min (that is, immediately after returning to OCV), sharp satellite features emerge around 743 eV between the M_4 and M_5 edges, along with a broader tail feature centred at 757 eV. After 20 min, the 735–743 eV region becomes flattened, and the 757 eV feature diminishes substantially, continuing to fade by 40 min. Notably, these spectral changes are observed in TEY mode but absent in the bulk-sensitive total fluorescence yield (TFY) mode (Fig. 4c), confirming that the spectral variations arise from the interfacial region rather than the bulk electrolyte.

The strong satellite features at 0 min reflect a pronounced interaction between Cs^+ and oxygen-containing species during catalysis. Within the NOLI pocket, several species could contribute to this interaction. First, Cs^+ may coordinate with the phosphonic acid groups of TDPA ligands, as supported by O K-edge data. Indeed, Cs–TDPA complexes display pronounced satellite features in their M-edge spectra (Supplementary Fig. 15). Second, although Cs– H_2O interactions typically do not give rise to sharp satellite features, our prior work on graphene–electrolyte interfaces shows that at extremely high interfacial Cs^+ concentrations ($\sim 8 \text{ M}$), satellite features can emerge due to the formation of Cs–anion pairs⁵⁶. Finally, given the applied bias and CO_2 -saturated environment, the observed features could originate from direct interactions between desolvated Cs^+ and surface adsorbates, for example, surface hydroxyl ($^*\text{OH}$), which has been recently reported³⁷, or reaction intermediates, particularly $^*\text{CO}_2^-$, as supported by SEIRAS data.

To further elucidate the origin of the satellite features, we conducted in situ Cs M-edge measurements during an online electrolyte exchange from CO_2 - to Ar-saturated conditions at OCV. During the exchange, we observed a gradual decline in the intensity of the M_4 and M_5 main edges (Supplementary Fig. 16), indicating a decrease in interfacial Cs^+ concentration. This result is corroborated by electrochemical capacitance measurements showing lower ion concentration under Ar versus CO_2 -saturated conditions (Supplementary Fig. 17). After the Cs signal stabilized, the same measurement protocol was repeated: preconditioning at $-0.2 V_{\text{RHE}}$ followed by spectral acquisition every 17 min. Under Ar, the Cs M-edge intensity did not recover to the level observed under CO_2 , confirming that CO_2 is required to maintain a high interfacial Cs^+ concentration. As shown in Fig. 4d and Supplementary Fig. 18, while a weak 757-eV satellite feature appeared at 0 min, its intensity was markedly lower than that under CO_2 -saturated conditions. Upon reintroduction of CO_2 , the satellite feature promptly recovered (Supplementary Fig. 19). Since Cs–ligand, Cs– HCO_3^- /Cs– CO_3^{2-} and Cs– H_2O interactions remain largely unchanged between Ar and CO_2 conditions, these observations point to interactions with CO_2 -derived surface intermediates as the origin of the satellite features. Alternative assignments—for instance, Cs– $^*\text{OH}$ interactions—can be ruled out, since under Ar conditions the coverage of $^*\text{OH}$ (if present) should be higher owing to the absence of competitive adsorption of CO_2 intermediates, and would thus be expected to produce a stronger, not weaker, satellite signal. Instead, the observed trend is fully consistent with a direct interaction between desolvated Cs^+ and CO_2 -derived intermediates, most probably adsorbed $^*\text{CO}_2^-$. DFT calculations further support this conclusion. The DOS of adsorbed $^*\text{CO}_2^-$ stabilized by a partially desolvated Cs^+ (Fig. 4e) shows noticeable orbital overlap between Cs 5p and O 2p states in the -6 to -12 -eV region, as well as additional Cs 5s–O 2s overlap below -20 eV. s orbital overlapping is negligible in salt crystals such as CsHCO_3 and Cs_2CO_3 , suggesting an even stronger Cs–O interaction in the adsorbed $^*\text{CO}_2^- \cdots \text{Cs}^+$ complex.

To mechanistically rationalize how desolvated Cs^+ promotes CO_2 activation, we generated seven idealized interfacial states (Fig. 4f) that discretize a desolvation coordinate from Cs^+ with six explicit H_2O molecules, $\text{Cs}^+(\text{6H}_2\text{O})$ (outer Helmholtz/diffuse layer-like environment)^{56,57} to $\text{Cs}^+(\text{1H}_2\text{O})$ (extreme near-surface coordination). These states are thermodynamic sampling points rather than a presumed kinetic pathway. We employed grand canonical DFT (GC-DFT) calculations and evaluated a grand free energy of CO_2 adsorption ($\Delta\Omega$) referenced to the clean Ag(111) surface with a solvated Cs^+ , and the Gibbs free energy of solvated CO_2 . Details can be found in the Methods. We used hydrated $\text{Cs}^+(\text{6H}_2\text{O})$ far from the interface as the reference point, where a physisorbed, nearly linear CO_2 (O–C–O 179.9° , average Cs–O $4.56 \text{ \AA}$) is observed. Slight disruption (lower symmetry) of the solvation shell ($6\text{H}_2\text{O}$) begins to bend the CO_2 (angle $\rightarrow 171.0^\circ$) at the Ag surface (Fig. 4g). Progressive water removal ($5 \rightarrow 1\text{H}_2\text{O}$) shortens Cs–O (of CO_2) from $3.51 \rightarrow 2.83 \text{ \AA}$ and CO_2 –Ag contact from $2.82 \rightarrow 2.19 \text{ \AA}$,

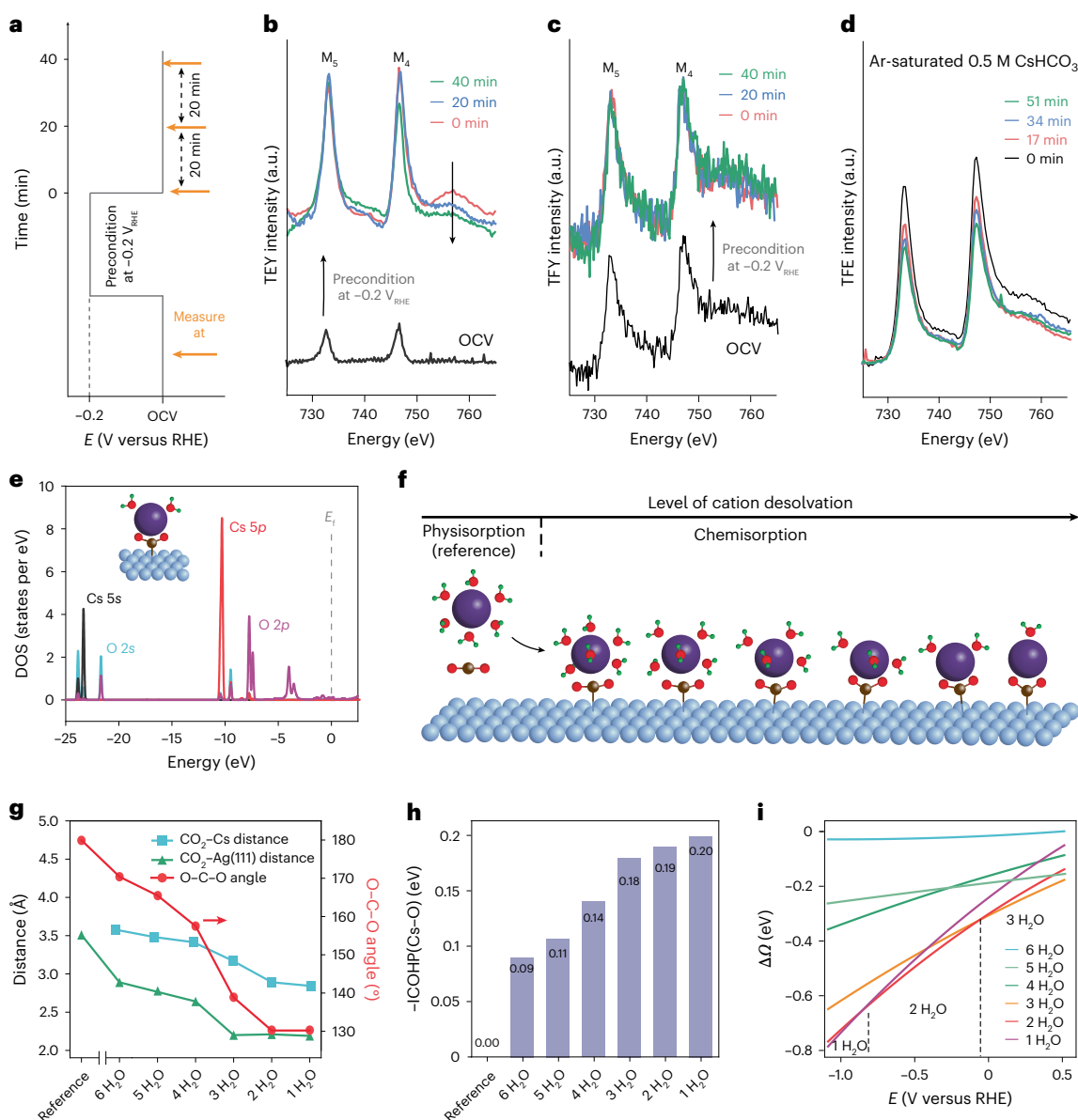


Fig. 4 | Cs- $^{*}\text{CO}_2$ interaction probed by in situ XAS and DFT calculation.

a, A schematic of the experimental protocol. XAS spectra were collected before preconditioning, and at 0 min, 20 min and 40 min after preconditioning at $-0.2\text{ V}_{\text{RHE}}$. **b, c**, In situ Cs M-edge spectra acquired in TEY mode (**b**) and TFE mode (**c**) in 0.5 M CO_2 -saturated CsHCO_3 . **d**, In situ Cs M-edge spectra acquired in TEY mode in 0.5 M Ar -saturated CsHCO_3 . **e**, The PDOS of an adsorbed $^{*}\text{CO}_2$ stabilized by a partially desolvated Cs^+ ; O orbitals are selected exclusively from the two O atoms of adsorbed $^{*}\text{CO}_2$. **f**, GC-DFT-optimized adsorption structures of CO_2

under varying degrees of Cs^+ desolvation. Colour code: Cs (purple), O (red), C (grey) and H (white). **g**, Structural descriptors including average Cs-O- CO_2 (light blue), Ag(111)- CO_2 (green) bond lengths and the O-C-O bond angle (red) of adsorbed CO_2 across the configurations in **f**. **h**, Negative ICOHP values reflecting the bond strength between Cs and CO_2 in the structures shown in **f**. **i**, The calculated potential-dependent $\Delta\Omega$ stability of different $^{*}\text{CO}_2$ -Cs structures in **f** with varying degrees of cation desolvation.

while $\text{ICOHP}(\text{Cs}-\text{O})$ magnitudes become more negative (-0.11 , -0.14 , -0.18 , -0.19 and -0.20 eV for 5, 4, 3, 2 and $1\text{ H}_2\text{O}$, respectively) (Fig. 4h), evidencing an increasing partial covalent component in Cs-O bond. This enhanced interaction yields a more activated $^{*}\text{CO}_2$ intermediate that is structurally predisposed for downstream protonation (Fig. 1f).

Potential-dependent $\Delta\Omega$ profiles (Fig. 4i) show several crossovers: Above $-0.1\text{ V}_{\text{RHE}}$, $\text{Cs}^+(3\text{ H}_2\text{O})$ minimizes $\Delta\Omega$; between $-0.1\text{ V}_{\text{RHE}}$ and $-0.8\text{ V}_{\text{RHE}}$ a ($2\text{ H}_2\text{O}$) state becomes competitive, and more negative than $-0.8\text{ V}_{\text{RHE}}$ the ($1\text{ H}_2\text{O}$) state is lowest. These results suggest that under increasingly reductive conditions, activated CO_2 is preferentially stabilized by Cs^+ with progressively higher degrees of desolvation. Since such desolvated cation-intermediate interactions are thermodynamically promoted by applied bias, we propose that they are not unique

to ligand-defined NOLI interfaces but represent a general feature of electrochemical solid-liquid interfaces once cation desolvation can be sustained.

In conventional ligand-free metal electrodes operated in dilute aqueous electrolytes, cations are expected to remain largely solvated, limiting the magnitude and persistence of such specific interactions. This provides a natural explanation for why cation effects are commonly observed on bare metal surfaces yet are typically more modest and highly sensitive to electrolyte identity and operating conditions. By contrast, the NOLI architecture stabilizes a locally ion-rich, water-deficient interfacial microenvironment, thereby amplifying the population of desolvated Cs^+ ions and rendering their direct interaction with $^{*}\text{CO}_2$ experimentally observable and catalytically important.

More broadly, similar desolvation-enabled cation–intermediate interactions may be accessible in other confined or low-water-activity interfacial environments, such as highly concentrated electrolytes³⁶, ionic-liquid-containing media⁵⁸ and nanoporous or ionomer-containing electrodes^{59,60}, where ion crowding and restricted solvation are more likely to occur. Altogether, these insights underscore the importance of interfacial design—specifically, engineering ion-rich, water-deficient microenvironments—as a general strategy to harness desolvated cation–intermediate interactions for enhancing reactivity across a wide range of electrochemical transformations.

Conclusion

We combined in situ vibrational and X-ray spectroscopic techniques with first-principles simulations to elucidate the solvation structure and interfacial behaviour of Cs^+ under CO_2 -reducing conditions. SEIRAS measurements captured the full evolution of the Cs^+ solvation environment under applied bias, revealing that within the catalyst–electrolyte interface confined by a detached ligand layer, desolvated cations facilitate CO_2 activation at unusually low overpotentials. This phenomenon directly accounts for the enhanced CO_2R activity observed in this NOLI system. In situ XAS measurements further revealed that the promotional effect of desolvated cations originates from direct interactions between Cs^+ and surface-bound intermediates. Notably, DFT-assisted XAS analysis showed that the interaction between desolvated Cs^+ and $^*\text{CO}_2^-$ species possesses partial covalent character, demonstrating that cations can stabilize intermediates, and thus modulate electrochemical reactions through mechanisms beyond pure electrostatic stabilization.

Altogether, this work provides direct spectroscopic evidence of desolvated Cs^+ under operando conditions and clarifies its role in stabilizing $^*\text{CO}_2^-$ intermediates, enabling efficient CO_2 -to- CO conversion at low overpotentials. Moreover, we highlight that desolvated cations can engage in orbital hybridization with key intermediates, thereby influencing reaction kinetics through direct electronic effects. Our findings highlight that ligand-realized, ion-rich interfacial confinement is a powerful and generalizable design principle for catalytic systems seeking to harness and amplify cation effects. This strategy opens avenues for microenvironment engineering to optimize activity and selectivity across a wide range of electrochemical reactions.

Methods

Synthesis of Ag nanoparticles

Silver nanoparticles were synthesized following previously reported procedures¹⁹. In a three-neck flask, 10 ml of trioctylamine (Sigma-Aldrich, 98%) was degassed with nitrogen at 130 °C for 30 min. After cooling to room temperature, 0.50 mmol of silver(I) acetate (Sigma-Aldrich, 99.99%) and 0.25 mmol of TDPA (Sigma-Aldrich, 98%) were added. The mixture was heated at 130 °C for 1 h under a nitrogen atmosphere. Upon cooling to 50 °C, the reaction mixture was transferred to a centrifuge tube, and 35 ml of ethanol was added to precipitate the nanoparticles. After centrifugation at 4,000g for 15 min, the nanoparticles were redispersed in 10 ml of hexane. To further narrow the particle size distribution, acetone (~10 ml) was added dropwise to induce turbidity, followed by centrifugation at 16,100g for 10 min. The final nanoparticles were redispersed in hexane for storage.

Preparation of Cs–TDPA complexes

TDPA was dissolved in ethanol to a final concentration of 50 mM. CsOH powder was gradually added to the solution until the Cs^+ concentration reached 0.1 M. Dissolved TDPA and CsOH are expected to undergo an acid–base reaction, forming the Cs–TDPA compound, where oxygen atoms of TDPA directly interact with Cs cations.

Electrochemical measurements

All electrochemical measurements were performed in a custom-built H-cell. The Ag/AgCl (3 M KCl, World Precision Instruments) electrode

was used as the reference, and a platinum wire served as the counter electrode. The potentiostat (Biologic) automatically compensated for 85% of the solution resistance; the remaining 15% was manually corrected in post-processing. For catalyst preparation, 29 μg of Ag nanoparticles were dropcast onto a 1- cm^2 carbon paper electrode (Toray 29AA).

Gaseous reaction products were analysed using a gas chromatograph (SRI Instruments) equipped with both thermal conductivity and flame ionization detectors. Product-specific FEs were determined by converting the quantified molar production rates into charge equivalents and normalizing to the total charge passed during electrolysis. To reliably quantify the FE values at low current densities, the surface area of the working electrode was increased, and the CO_2 flow rate was reduced to 5 sccm when total current is below 0.5 mA (compared with 20 sccm used at higher current densities), enabling gas products to accumulate to higher concentrations before gas chromatograph analysis and ensures that detected signals remain well above the instrument's quantification limit.

The electrochemical surface area (ECSA) of the Ag NOLI catalysts was quantified using the Pb underpotential deposition method, following a procedure that was reported previously¹⁹. In brief, cyclic voltammetry was performed between –0.10 and –0.45 V versus Ag/AgCl (1 M KCl) at a scan rate of 50 mV s^{-1} in an electrolyte containing 3 mM $\text{Pb}(\text{ClO}_4)_2$, 10 mM HClO_4 and 100 mM NaClO_4 . A polished Ag foil with a geometric area of 2 cm^2 was used as the reference, and its Pb stripping charge (361.7 $\mu\text{C cm}^{-2}$) was taken as the standard for calculating the ECSA of the Ag NOLI catalysts.

Electrochemical impedance spectroscopy measurements were performed to quantify the electrochemical double-layer capacitance of the Ag NOLI electrode. All electrochemical impedance spectroscopy experiments were conducted in 0.1 M CsHCO_3 electrolyte. Impedance spectra were collected by applying a 10-mV AC perturbation over a frequency range from 100 kHz to 100 mHz, with eight data points per decade. Each measurement was repeated three times to ensure reproducibility. The resulting spectra were analysed using an equivalent circuit consisting of a solution resistance (R_s) in series with a parallel combination of a constant phase element (CPE; Q) and charge-transfer resistance (R_{ct}). CPE was used in place of an ideal capacitor to account for non-ideal capacitive behaviour of the electrochemical double layer⁶¹. The fitted CPE exponent (α) was greater than 0.9 across all measurements, indicating near-ideal capacitive response. Accordingly, the Q values were taken as close approximations of the double-layer capacitance (C_{dl}).

Electrode preparation for SEIRAS

The reflecting surface of a 60° Si prism (Pike Technologies) was polished sequentially with 6- μm , 1- μm and 0.25- μm diamond pastes for 10 min each to expose a fresh Si surface. The prism was then rinsed with deionized (DI) water and sonicated in DI water, acetone and DI water for 10 min each. After cleaning, the prism surface was etched in 5 ml of 40 wt% NH_4F solution at 25 °C for 2 min to achieve a hydrogen-terminated surface, followed by thorough rinsing with DI water.

Gold films were deposited by immersing the etched Si prism in a gold plating solution at 60 °C for 200 s. The plating solution was prepared by adding 200 μl of 49% HF to a 10-ml aqueous mixture containing 10 mM $\text{NaAuCl}_4 \cdot 2\text{H}_2\text{O}$, 100 mM Na_2SO_3 , 33 mM $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$ and 33 mM NH_4Cl . To improve film adhesion and uniformity, the freshly deposited gold layer was first removed using aqua regia, followed by re-plating until a smooth, mirror-like surface was obtained. A high-quality gold film exhibited a resistance less than 10 Ω .

Before catalyst deposition, the Au film was electrochemically activated by cycling the potential between 0.2 V and 1.5 V (versus 3 M Ag/AgCl) in 0.1 M H_2SO_4 for three cycles. This step is critical for cleaning the surface and enabling sufficient surface enhancement.

The activated Au films were rinsed with DI water and dried under nitrogen. For Ag NP measurements, 35 μg of Ag nanoparticles were dropcast onto the Au-coated prism. For Ag film control samples, a 50-nm Ag film was deposited directly onto the activated Au surface by thermal evaporation.

In situ SEIRAS measurements

The catalyst-loaded Si prism was assembled into a custom electrochemical cell and mounted onto a VeeMax III ATR accessory (Pike Technologies). IR spectra were collected using a nitrogen-purged Bruker Vertex 70 FTIR spectrometer equipped with a mercury cadmium telluride detector. Spectra were recorded with a 4 cm^{-1} resolution, 32-scan averaging and a scan velocity of 20 kHz during linear sweep voltammetry (2 mV s^{-1}). Each spectrum acquisition took approximately 50 s, and the reported potential corresponds to the end of the acquisition period. All SEIRAS spectra are reported as absorbance, calculated using

$$\text{Absorbance} = -\log_{10} \frac{S}{R}, \quad (2)$$

where S is the single-beam sample spectrum and R is the single-beam reference spectrum.

STEM measurement

Electrochemical scanning transmission electron microscopy (EC-STEM) and four-dimensional scanning transmission electron microscopy (4D-STEM) were collected on the FEI ThemIS instrument at the molecular foundry, LBNL. The liquid-cell setup consisted of a Protochips Poseidon AX holder and two silicon chips with silicon nitride windows sandwiching a 500-nm electrolyte layer. The bottom chip features a three-electrode setup with glassy carbon working electrode, Pt counter electrode and Pt pseudo-reference electrode. Ag NOLI was dropcast onto the glassy carbon working electrode. The beam current was set to <1 pA to avoid beam damage, and the convergence angle was 0.46 mrad. 4D-STEM was collected on a Gatan K2 camera in situ mode. Data were processed using the py4DSTEM python library⁶².

XAS measurement

In situ XAS experiments were conducted at three synchrotron facilities: beamline 8.0.1.4 and 9.0.1 of the Advanced Light Source (ALS), Lawrence Berkeley National Laboratory (USA), and beamline B07 at the Diamond Light Source (UK). At ALS, the measurements were performed using a custom-built liquid flow cell featuring an ultrathin Si_3N_4 membrane (100-nm thickness), which serves as the X-ray transparent window and separates the liquid-phase electrochemical environment from the ultrahigh vacuum of the beamline. To ensure conductivity, the membrane was coated with 10 nm of Pt. The electrochemical setup inside the cell included a calibrated Ag/AgCl (3 M KCl) reference electrode and a platinum wire as the counter electrode.

At the Diamond Light Source, an in-house electrochemical flow cell⁶³ was employed. This setup included a platinum wire counter electrode and an Ag/AgCl-coated silver wire as the reference electrode. The continuous Si_3N_4 window (100 nm thickness) was coated with 2.5 nm of Cr as adhesive layer, a 10-nm layer of gold as current collector, followed by three layers of graphene. The quality and coverage of the graphene layer were verified by scanning electron microscopy and Raman spectroscopy to ensure minimal defects and uniform coverage.

For in situ measurements, the Ag NOLI sample was prepared by dropcasting Ag nanoparticles onto the Si_3N_4 membrane at a loading of 50 $\mu\text{g cm}^{-2}$. For Ag film control samples, a 2.5-nm chromium adhesion layer followed by a 15-nm silver film was deposited onto the Si_3N_4 window by thermal evaporation.

The XAS spectra of reference salts were measured on Beamline 9.3.2 of the ALS and B07 of the Diamond Light Source. Samples on 9.3.2 were dropcast and measured on a polished gold foil, all samples were dropcast as 3- μl droplets from a saturated solution and dried in a nitrogen box. A

platinum heater sample holder was used for temperature control and purified water for humidity control. The XPS capabilities of 9.3.2 were used to find a thin layer of salt at the outer edges of the coffee ring, to reduce charging and solid-state effects. Gold was used as a current collector for the drain current measurements. Therefore, the thermocouple was connected to a SR560 preamplifier. To verify that the observed Cs–O interactions are not an effect of Cs–electrode interactions, or background drift from the drain current measurement, the XPS analyser was employed to simultaneously measure the secondary electron yield, by collecting low-energy electrons with a kinetic energy of 40 eV. Reference measurements at the Diamond Light Source were performed on a p-doped Si wafer. Dropcasting was performed in a similar manner to the 9.3.2 samples. Fluorescence yield and drain current measurements were employed at the same time to verify that the observed spectral features are not an effect of electron-based XAS detection methods.

DFT details

All periodic DFT calculations were performed with the Vienna Ab initio Simulation Package⁶⁴ using the projector-augmented wave method⁶⁵ and the PBE exchange–correlation functional within the generalized gradient approximation, augmented by the dDSC dispersion correction to include van der Waals interactions^{66,67}. A plane-wave kinetic-energy cutoff of 400 eV was used in all stages (converged to less than -1 meV per atom in test calculations). Gas-phase and bulk structural relaxations were first carried out in vacuum until the maximum residual force fell below 0.01 eV $\text{\AA}^{-1}$ and the total energy change was <10⁻⁶ eV. Subsequent implicit-solvation optimizations employed VASPsol^{68,69} with force and energy convergence criteria of 0.02 eV $\text{\AA}^{-1}$ and 10⁻⁶ eV, respectively.

For bulk crystals (Cs_2CO_3 , CsHCO_3 and CsI) the Brillouin zone was sampled with a Γ -centred $5 \times 5 \times 5$ Monkhorst–Pack grid during geometry optimization. Surface models used a three-layer Ag(111) (4×4) supercell (bottom layer fixed), a 15 $\text{\AA}$ vacuum gap along z and a $5 \times 5 \times 1$ k -point grid for relaxation. For implicit-solvation and GC-DFT treatments, the simulation cell height was extended to 60 $\text{\AA}$ to host the dielectric continuum while retaining the atomically explicit slab and adsorbates; slab symmetry operations were preserved where applicable.

High-accuracy single-point calculations for projected density of states (PDOS), Bader charge analysis, differential charge density ($\Delta\rho$), ELF and bonding (COHP) metrics used denser Γ -centred meshes of $8 \times 8 \times 8$ (bulk) and $8 \times 8 \times 1$ (slabs). COHP and integrated -ICOHP values were obtained with LOBSTER (pbeVasFit2015 basis)⁷⁰ after verifying a charge spilling <2%. Bader charges were computed on the all-electron charge density reconstructed from the projector-augmented wave potentials with a fine FFT grid (spacing ≤ 0.10 $\text{\AA}$). Differential charge densities were generated as $\rho_{\text{total}} - \Sigma \rho_{\text{frag}}$ (identical supercells, fragment geometries frozen to the combined state). ELF was evaluated on the self-consistent real-space grid and analysed via selected z slices and line profiles along Cs–O (or Cs–intermediate) axes.

GC-DFT calculations

The grand canonical free energy was evaluated by GC-DFT calculations, which is a surface charging technique. Details can be found in our previous work, and here we summarize the key points^{71,72}. The net charge of the surface n surface is obtained as

$$n_{\text{surface}} = N_{\text{surface}} - N_{\text{surface,neutral}} \quad (3)$$

where N_{surface} and $N_{\text{surface,neutral}}$ are the number of electrons on the surface and the number of electrons in the neutral state. DFT energy for the charged surface is obtained as

$$E_{\text{surface}} = E_{\text{surface,raw}} + \epsilon_{\text{fermishift}} n_{\text{surface}} \quad (4)$$

where $E_{\text{surface,raw}}$ is the raw electronic energy of the surface and $\epsilon_{\text{fermishift}} n_{\text{surface}}$ is the correction term accounting for the difference ($\epsilon_{\text{fermishift}}$) in the reference energy of the electron between the internal

reference level and vacuum. Then, the grand canonical electronic energy of a surface model, $\Omega(U)$ is obtained as

$$\Omega(U) = E_{\text{surface}} - n_{\text{surface}} \mu_{\text{electron}} \quad (5)$$

where μ_{electron} is the chemical potential of an electron, which is defined as

$$\mu_{\text{electron}} = qU_{\text{vac}} = -eU_{\text{vac}}, \quad (6)$$

where U_{vac} is the potential of the system with reference to the vacuum level and q is the charge of an electron. The potential of the system with reference to the vacuum can be determined by two components, the Fermi level (E_F) with reference to the internal zero energy reference and the Fermi shift, which is the difference between the internal energy reference and the vacuum level

$$-eU_{\text{vac}} = E_F + E_{\text{fermishift}}. \quad (7)$$

For the metallic systems, the potential-dependent grand canonical energy, $\Omega(U)$, exhibits a quadratic behaviour around the potential of zero charge (U_0) in the vacuum scale

$$\Omega(U) = \Omega(U_0) - \frac{1}{2}C(U - U_0)^2 \quad (8)$$

where C is the capacitance of the surface. The potential of the system with respect to the standard hydrogen electrode (SHE) can be converted from U_{vac} as

$$U_{\text{SHE}} = U_{\text{vac}} - 4.44. \quad (9)$$

The linearized Poisson Boltzmann implicit solvation model implemented in VASPsol⁶⁹ is used to represent the polarizable electrolyte region. The dielectric constant of water, 78.4, and the Debye screening length corresponding to a 0.1 M concentration of electrolytes, 9.6 Å, were used. The surface slab is symmetrized along the z axis to avoid asymmetric potential in the implicit solvation region. Here, the implicit solvent thickness is set to 60 Å for the symmetrized slab.

Grand canonical free energy calculations of CO₂ adsorption

To evaluate the thermodynamic stability of *CO₂ adsorption configurations with different degrees of Cs⁺ desolvation under applied potential, we employed GC-DFT. For each configuration containing adsorbed *CO₂, an interfacial Cs⁺ cation, and n explicitly modelled water molecules on Ag(111), we define the grand free energy of CO₂ adsorption at electrode potential U (versus RHE) as

$$\Delta\Omega(U) = G(U)_{\text{Ag111+CO}_2+\text{Cs}^{+1}+(\text{H}_2\text{O})_n} - G(U)_{\text{Ag111+Cs}^{+1}+(\text{H}_2\text{O})_n} - G_{\text{CO}_2}, \quad (10)$$

where $G(U)_{\text{Ag111+CO}_2+\text{Cs}^{+1}+(\text{H}_2\text{O})_n}$ is the Gibbs free energy at electrode potential U of the Ag(111) slab bearing an adsorbed *CO₂, an interfacial Cs⁺ and n explicit water molecules, which is referenced to a clean Ag(111) surface with an adsorbed solvated Cs⁺ with n explicit water molecules, $G(U)_{\text{Ag111+Cs}^{+1}+(\text{H}_2\text{O})_n}$, and the free energy of solution-phase CO₂, G_{CO_2} (approximated as gas-phase CO₂ with one-half of its gas-phase entropy to mimic dissolution).

Data availability

Data supporting the findings of this study are available within the Article and its Supplementary Information. Atomic coordinates of the structures used in our DFT calculation are provided in Supplementary Data 1. Other relevant data are available from the corresponding authors upon reasonable request.

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Author contributions

Y.S., M.J. and J.F. conceived and designed the project under the guidance of M.B.S. and P.Y. Y.S. synthesized the materials and carried out the in situ SEIRAS studies with assistance from T.W., M.V.F.G. and A.S. Y.S., M.J. and J.F. performed in situ XAS measurements at the Advanced Light Source and analysed the data with assistance from L.J.F., J.L., H.D., S.N., X.Z. and J.G. M.J. and J.F. additionally conducted and analysed in situ XAS measurements at Diamond Light Source. M.J. carried out XAS measurements on the calcinated and humidified references at beamline 9.3.2 of the Advanced Light Source. Y.G., R.M.K. and P.S. carried out the DFT calculations. All authors contributed to the discussion of results and to the preparation of the paper.

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Competing interests

The authors declare no competing interests.

Additional information

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